Unit: 6 Unit 2: Exponents Radicals **Name:** ____________

Topic: Exponents **Date:** __________

Advanced (Habanero)

Q1. Simplify $2^3 \cdot 2^4$
(A) 2^5
(B) 2^6
(C) 2^7
(D) 2^8
(E) 2^{12}

Q2. Simplify $(3^2)^3$
(A) 3^5
(B) 3^6
(C) 3^7
(D) 3^8
(E) 3^9

Q3. Simplify $\frac{4^3}{4^2}$
(A) 4^1
(B) 4^2
(C) 4^3
(D) 4^4
(E) 4^5

Q4. Simplify $\frac{2^6}{2^4}$
(A) 2^1
(B) 2^2
(C) 2^3
(D) 2^4
(E) 2^5

Q5. Simplify $(2^2 \cdot 3^2)^3$
(A) $2^5 \cdot 3^5$
(B) $2^6 \cdot 3^6$
(C) $2^7 \cdot 3^7$
(D) $2^8 \cdot 3^8$
(E) $2^{12} \cdot 3^{12}$

Q6. Simplify $\frac{(2^3)^2}{2^4}$
(A) 2^1
(B) 2^2
(C) 2^3
(D) 2^4
(E) 2^5

Q7. Simplify $(3^2)^2 \cdot 3^3$
(A) 3^5
(B) 3^6
(C) 3^7
(D) 3^8
(E) 3^9

Q8. Simplify $\frac{5^4}{(5^2)^2}$
(A) 5^0
(B) 5^1
(C) 5^2
(D) 5^3
(E) 5^4

Open Ended Questions (FRQ)

Q9. Simplify the expression $\frac{(2^3 \cdot 3^2)^2}{(2^2 \cdot 3^3)^2}$

Q10. Simplify the expression $(2^2 \cdot 3^3)^3 \cdot \frac{2^4}{3^2}$

Chef's Tips

- Provide examples of different exponent rules
- Explain the product, quotient, and power rules for exponents
- Use real-world applications to illustrate the importance of exponents